

Experiment 8

1. Experiment Title

Access and Validate Form Fields Using Events

2. Software/Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript
- Live Server Extension

3. Theory

Form Field Access

JavaScript can access HTML form elements using DOM methods such as:

```
document.getElementById()
```

For example:

```
const nameInput =  
    document.getElementById("name");
```

This allows JavaScript to read and validate the value entered by the user.

Events

An event is an action that occurs on a webpage. JavaScript can respond to events using `addEventListener()`.

The experiment demonstrates:

- **input** – performs validation while the user types.
- **change** – detects changes in select, date, and checkbox fields.
- **blur** – validates a field when the user moves away from it.
- **submit** – validates the complete form before submission.

Example:

```
emailInput.addEventListener(  
    "input",  
    validateEmail  
);
```

Regular Expression

Regular expressions are used to validate email addresses and phone numbers.

Example:

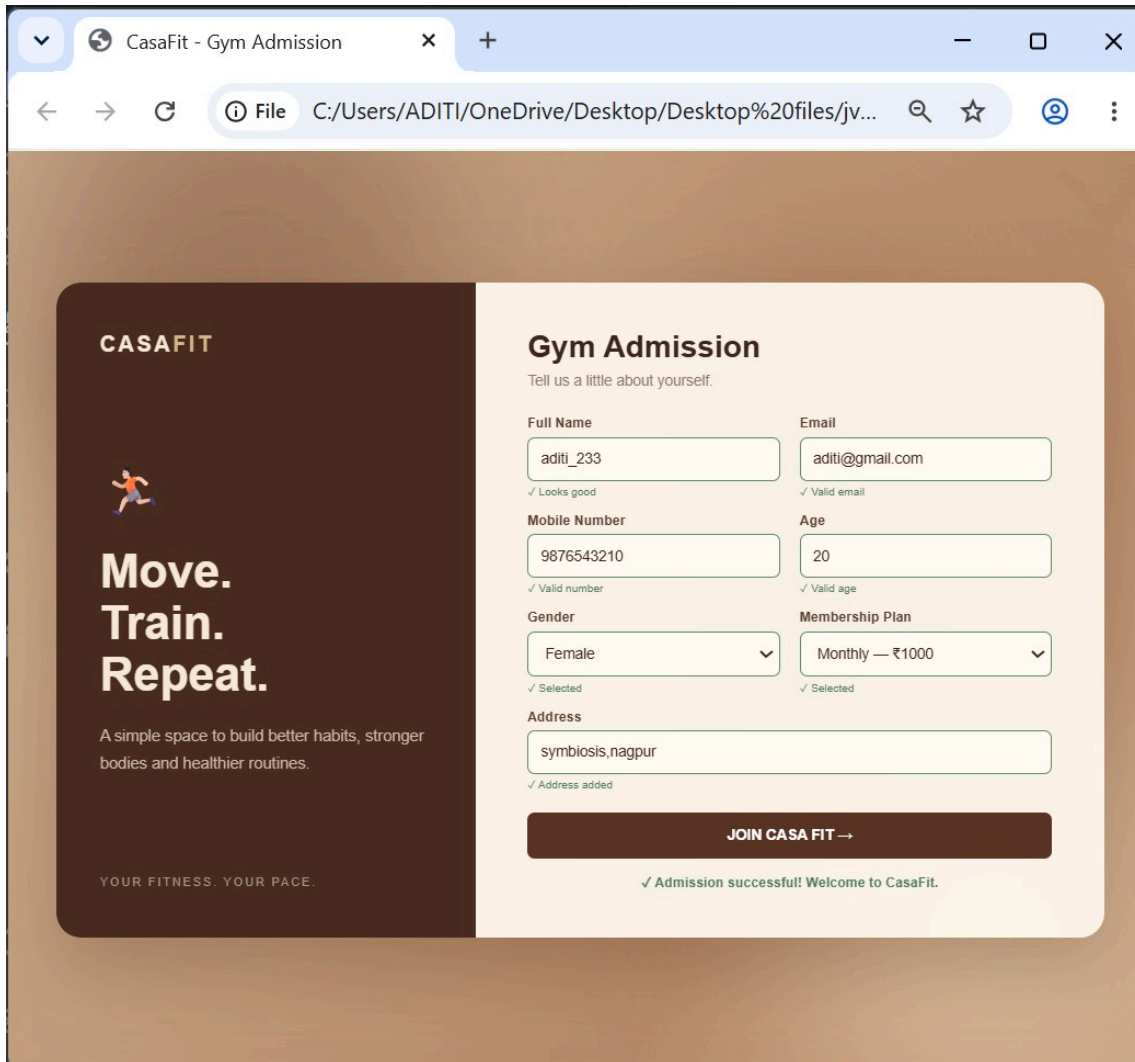
```
let pattern = /^[6-9]\d{9}$/;
```

This checks whether a mobile number contains 10 digits and begins with a valid digit.

Live Validation

Live validation provides immediate feedback to the user. Valid fields are displayed with a success indication, while invalid fields display an error message.

4. Output



5. Result / Conclusion

The **Gym Admission Form** was successfully designed and implemented using HTML, CSS, and JavaScript. JavaScript DOM methods were used to access form fields, while `input`, `change`, `blur`, and `submit` events were used for real-time validation. Regular expressions were used for email and phone-number validation. The system provides immediate feedback and prevents submission when invalid information is entered.